A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Σn(n - 1) = (40 x 39) + (30 x 29) + (20 x 19) + (10 x 9) = 1560 + 870 + 380 + 90 = 2900 **1** N(N - 1) = 100 x 99 = 9900 **1** D = 9900 / 2900 = 3.41 (accept 3.4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B **1** The index does not identify which species are present / a rare or important species may be missed – depends on identification skill and sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
02.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Volume of ash is much less than the original waste (less landfill) **1** Heat used for district heating / electricity generation (energy recovery) **1** No sorting / complicated management needed **1** Destroys pathogens / flammable organic wastes converted to CO2 and water **1** Refuse-derived fuel can replace fossil fuels (eg cement kilns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
02.2	**1 mark each** 1 mark for each of the 4 points below. **1** High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) **1** Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone **OR** removes heavy metals for storage at high pH **1** Wet waste needs extra fuel to burn and lowers combustion temperature – composting recovers its nutrient value **1** Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ash may contain concentrated heavy metals / toxic residues **1** Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
02.4	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Fuel (natural gas) needed to burn wet / non-flammable wastes **1** Cost of treating effluent gases (SO2, NOx, HCl, smoke) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
03.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Disposal of items containing mercury:** batteries, thermometers, fluorescent lights **1** Chemical plants producing chlorine using mercury electrodes **1** Combustion of coal **1** Gold mining (amalgamation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

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Question	Marking guidance	Marks
03.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) **1** Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters **1** Disposal / storage of wastes at high pH to reduce solubility **1** Encapsulation of mercury-containing hazardous waste in cement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are more soluble at low pH / less soluble at high pH **1** So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** The effects of two pollutants interact to produce a different (greater) effect than either alone **1** Cadmium and zinc – ozone and sulfur dioxide **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium **1** Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet **1** International agreement on ownership of the seabed and its resources is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Collectors disturb the seabed and kill benthic organisms living on it **1** Separating nodules from sediment creates plumes that increase turbidity with unknown consequences **1** Re-deposition of sediment covers and kills filter feeders and other benthic organisms – slow recovery in the deep sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) **1** Exploitable land deposits are limited and mainly in China – chemical separation is complex and produces large amounts of toxic waste **1 Bacterial adsorption:** dissolved rare earth ions adsorb onto the cell surfaces of bacteria (Bacillus subtilis) and are washed off at specific pHs to separate them **1** Deep sea sediments contain relatively high concentrations but no viable extraction method yet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
06.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up **1** High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) **1** Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) **1** Birds of prey are at the top of the food chain so receive the highest concentrations **1** Sub-lethal doses caused thin eggshells that broke / reduced breeding success – higher doses caused sterility or death **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **trophic level** → the feeding level in a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly **1 Disadvantage:** high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase) – chronic effects (memory loss, depression) – suspected carcinogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	2
06.3	1 mark for the point below. **1** Pyrethroids are highly toxic to fish (even though mammal toxicity is low) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) **1** They are moderately persistent so remain in the soil long enough to be absorbed **1** Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats **1** So bees / pollinators are less exposed than when the crop is sprayed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Total = 42 + 26 + 31 + 9 + 22 = 130 kW **1** 31 / 130 x 100 = 23.8% (accept 24%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 3 points below. **1 Cavity wall insulation:** the foam or fibre traps air in small pockets so convection in the cavity is prevented and the low thermal conductivity of still air reduces conduction through the wall **1 Low-emissivity double glazing:** the still air or inert gas between the panes conducts heat poorly, and the low-emissivity coating reflects infrared radiation back into the room instead of letting it radiate out **1 Heat exchanger:** outgoing stale warm air passes next to, but does not mix with, incoming cold fresh air, so most of the heat is transferred to the incoming air and the building can be ventilated without losing that heat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each extra layer saves less heat than the one before, because the layers already in place carry most of the temperature difference, so the saving per unit thickness falls **1** Beyond a certain thickness the cost and the embodied energy of making and fitting more insulation are never repaid by the energy it saves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** A large power station runs at higher temperatures and pressures and uses large efficient turbines, so it converts a greater proportion of the fuel energy into electricity **1** The heat rejected by a large station is low-grade and the station is usually too far from any user, so it is lost to a river or cooling tower **1** A CHP unit is in the building that needs the heat, so the heat rejected by its generator is piped straight to the heating and hot water, and 70 - 80% of the fuel energy is used rather than about 35% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **combined heat and power** → using the waste heat from generation for heating	3

Do not write outside the box

Question	Marking guidance	Marks
07.5	**Any 5 of 6** 1 mark each for any 5 of the 6 points below, in any order, maximum 5. **1** Occupancy sensors and programmable thermostats so that lighting and heating run only where and when people are present **1** LED lighting, which converts a far greater proportion of the electricity into light and much less into waste heat than filament lamps **1** Large south-facing windows and a thermally massive structure to make use of passive solar gain, with shading to prevent overheating in summer **1** Heating water on demand at the point of use, so no energy is wasted keeping a tank hot and none is lost in the pipe run **1** Choosing materials with low embodied energy, and re-using the existing structure rather than demolishing and rebuilding **1 Changes in behaviour:** turning equipment off rather than leaving it on standby, closing doors, and setting the thermostat a degree lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-235; AQA spec 3.3.4.2.2-3.3.4.2.3.	
08.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Deforestation:** less interception so less evaporation from leaves – less transpiration so less water returned to the atmosphere and less precipitation downwind – more infiltration and surface runoff, higher river flows / flooding **2 Urban development:** impermeable concrete / tarmac reduce infiltration and aquifer recharge – runoff is faster and greater, increasing river flooding downstream – drains carry water quickly to rivers **2 Agriculture:** soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff – irrigation increases evaporation – loss of soil biota (worms) reduces infiltration and water retention – drainage lowers water tables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **infiltration** → water soaking into the soil surface • **impermeable** → does not let water through **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	6

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation **1** More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods **1** Changed wind patterns move humid air to different areas **1** Sea level rise / salt incursion into coastal aquifers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity **1** Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
09.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 High-level:** used fuel rods / fission products separated in reprocessing – vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) **2 Intermediate:** fuel cladding / metal tubes around fuel rods, filters from reprocessing – mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults **2 Low-level solid:** contaminated clothing, equipment – sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radioactive decay of the concentrated fission products generates heat which could damage containers **1** Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1 Liquids:** filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) **1 Gases:** filtered (HEPA / charcoal filters) to remove particles and iodine before release **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles:** dB is logarithmic (10 dB = 10x; 3 dB = 2x) so small reductions in dB are large reductions in intensity – effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • **Aircraft design:** high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • **Aircraft operation:** take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • **Airport design / location:** away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • **Railway:** track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • **Road:** porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • **Industrial:** silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • **Domestic:** acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • **Evaluation:** which approaches are most effective / cost-effective – conflicts (airports near cities for convenience) – legislation and monitoring (Leq 57 dB contours) needed to enforce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• Any appropriate named example, including ones that are not listed here. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Because the decibel scale is logarithmic, a 10 dB cut is a tenfold reduction in intensity and 3 dB halves it, so modest-sounding design changes matter. Aircraft engines have been made quieter by high-bypass-ratio turbofans, chevron nozzles, hush kits and acoustic liners, with fairings and composite airframes reducing airframe noise. Operation can be changed too: steeper take-off angles and flight paths that avoid housing, constant-descent approaches, restricted reverse thrust, night curfews, Chapter noise limits enforced by fines and quota counts. Airport location and layout - away from population centres, engine-test areas and taxiways kept from housing, acoustic walls and baffle mounds, land-use planning and free insulation for residents - deal with the noise that remains.	
	Railways use track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication on curves and composite brakes. Roads use porous or rubber-crumb asphalt, acoustic fences and embankments, tree planting, engine insulation and aerodynamics, traffic routing and double glazing. Industry fits silencers to compressors, presses rather than stamps, uses nylon rollers, drills instead of pile-driving, surrounds piling with bubble curtains to protect cetaceans, times blasting, hangs acoustic mats and operates remotely, with ear protection as a last line. Domestic appliances use acoustic absorbers and volume limiters.	
	Design changes at source are the most effective because they reduce noise for everyone; operational limits depend on enforcement; location works only for new developments, and airports are still sited near cities for convenience. Legislation and monitoring of noise contours (57 dB L_{eq}) are what turn these options into actual reductions.	
	Weaker answer - Level 2 (5 marks)	
	Noise pollution can be reduced in different ways. Aircraft can be designed with quieter engines and can fly at night less often. Airports can be built away from cities. Roads can have sound barriers next to them and quieter road surfaces. Trains can have their tracks polished and use better brakes. Factories can use quieter machines and workers can wear ear defenders. Noise can also be reduced by planting trees. These methods reduce the noise that reaches people.	
	What makes the difference	
	• Level 2 gives correct but generic examples ('quieter engines'); Level 3 names the design features (high bypass ratio, chevron nozzles, acoustic liners) the question's 'named examples' requires.	
	• Level 3 organises by design / operation / location as the question asks; Level 2 mixes them.	
	• The decibel-scale point and the closing evaluation of which approaches work best lift Level 3 into the top band.	
10.2	1 mark for the point below. **1** The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	
11.1	**Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why non-renewable:** formed too slowly – geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock – Lasky's principle – cut-off ore grade and its drivers • **Extraction technologies:** mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • **Reducing demand:** recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • **Environmental limits:** impacts of mining and their control – energy costs rising with lower grades – conflicts	25
	• **Judgement:** technology can extend supplies substantially but only within energy and environmental constraints – circular economy essential *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility – DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • **Reducing impact:** restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • **Alternatives:** cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • **Trade-offs:** yield, cost, human health (organophosphates), resistance evolution, pollinator services – risk:benefit – sustainability and the circular economy	25
	• **Judgement:** no pesticide is perfectly specific – combination of property-based selection and integrated control is most sustainable *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	

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